Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

FIFTH SEMESTER B.TECH. (ELECTRICAL &ELECTRONICS)

MAKE-UP EXAMINATIONS, DECEMBER 2024 - JANUARY 2025

SUBJECT: COMMUNICATION SYSTEMS [ELE 3126]

REVISED CREDIT SYSTEM

Time: 3 Hours **Date: 03 January 2025** **MAX. MARKS: 50**

Instructions to Candidates:

- Answer **ALL** questions.
- Missing data may be suitable assumed.

1A. For the amplitude modulated spectrum shown in figure ***(04)***

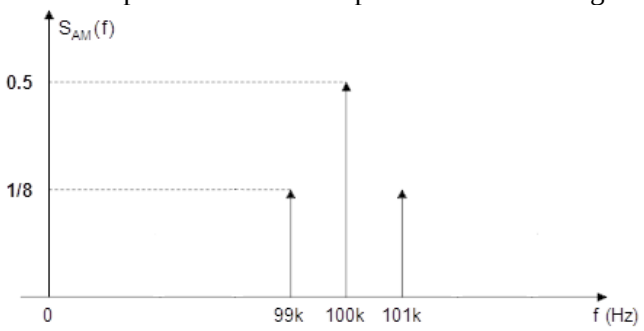


figure Qn(11)

i. Determine overall modulation index
ii. Determine power efficiency of modulation

1B. The input to a square law modulator is V_i given by V_i=10 cos 1000πt + 0.05cos200πt. Assume the characteristics of the nonlinear device used to be $V_0 = V_i + 0.1Vi^2$. Then it is passed through a bandpass filter (BPF) with frequency range of 400Hz to 600Hz. Compute the expression for the BPF output. ***(03)***

1C. A single tone FM is represented by the voltage equation as ***(03)***
$V(t) = 12\cos(6 \times 10^8 t + 5 \sin 1250t)$. Determine

i. Carrier frequency
ii. Modulating frequency
iii. Modulation index
iv. Frequency deviation
v. Highest and lowest frequencies obtained by the modulated signal.
vi. Power dissipated in 10Ω resistor.

2A. Describe the FM stereo multiplexing and demultiplexing technique with block diagram. ***(04)***

2B. An analog message signal is limited in its excursions to the range from -20 to +20 volts. If there are 4 quantization level, then draw the unipolar NRZ representation of the PCM output for the given sample values: ***(03)***

Sample values: 11.3 5.6 -1.7 -12.4

2C. 24 voice signals are sampled uniformly and then time division multiplexed. The sampling operation uses the flat-top samples with 1microsec duration. The multiplexing operation includes provision for synchronization by adding an extra pulse of sufficient amplitude and 1micro second. Assuming a sampling rate of 8KHz, calculate the spacing between successive pulses of the multiplexed signal. ***(03)***

3A. A direct sequence spread BPSK system uses three D flip flops and feedback taps to the modulo 2 adder is [1,3]. ***(04)***

a. Construct Linear Feedback Shift Register in the given configuration.

b. Determine the PN sequence with initial state 100.

c. Draw the DSSS-BPSK waveform for a data bit 0, in the given configuration.

3B. For a Binary Frequency Shift Keying technique , ***(03)***

a) Obtain the coordinates of message points and then draw the signal constellation diagram.

b) Draw the BFSK waveform for the message signal 011010 (consider bit rate = carrier frequency).

3C. In a digital communication system, the bit rate of NRZ signal is 1Mbps and the carrier frequency is 100MHz. Find the symbol rate of transmission and bandwidth requirement of the channel in the following cases: ***(03)***

a) BPSK system b) QPSK system

4A. A (7,4) linear block code has a generator matrix ***(03)***

$$G=\begin{bmatrix}1&1&0&1&0&0&0\\0&1&1&0&1&0&0\\1&1&1&0&0&1&0\\1&0&1&0&0&0&1\end{bmatrix}$$

a) Find the code words to be transmitted for the message, m= [1 1 0 1]

b) Find the parity bit representation as a linear combination of message

c) Check whether the transmitted code will receive correctly or not (by considering the orthogonal matrix of G as parity check matrix).

4B. If the received sequence is [011100], then using Viterbi algorithm, identify the message bits(decoded sequence), for a convolutional encoder with the given state diagram. ***(05)***

[where binary assignment of states are as given: a = 00, b = 10, c= 01 and d = 11]

Present state	Next State (with input=0)	Next State (with input =1)	Code Vector (with input=0)	Code Vector (with input=1)
A	A	B	00	11
B	C	D	01	10
C	A	B	11	00
D	C	D	10	01

4C. Let a wireless communication standard use slow FHSS with the given specifications. If the input binary sequence transmitted is 10110001 and the PN sequence generated is 0110. Sketch the frequency variation for one complete period of the PN sequence in slow FHSS. [Let the number of MFSK tones (M)= 4 and length of PN segment per hop(k) = 2, Assume that the carrier is hopped to a new frequency after transmitting two symbols] ***(02)***

5A. Assume that Bob sent a plain text "**webinarpostponed**" , to Alice by using the transposition cipher method for security purposes. The enciphering is performed using two steps. i) rail fencing by a depth of 4 and ii) column reading respectively. Column reading is done in the order of keystream as ***(04)***

Key : 4 2 3 1

Find the cipher text.

5B. For a smart irrigation system with 2 soil moisture sensors, 2 temperature sensors, a relay to switch no/off water pump and node MCU as the controller. Develop MQTT protocol architecture for IoT, with Node MCU as the gateway and mobile app for display. ***(03)***

5C. Let Ethernet follow a hexadecimal MAC address format. Show how the address 47:20:1B:2E:08:EE is sent out on ethernet cable and find the address type. ***(03)***